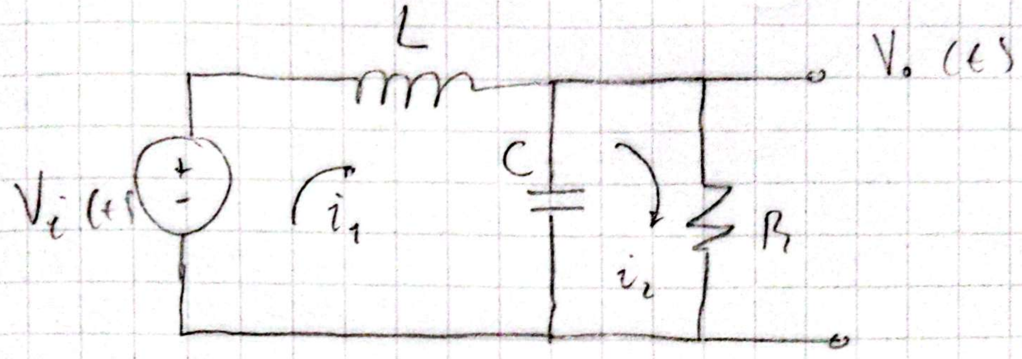

Malla 1:

$$V_i(t) = L\frac{di_1(t)}{dt} + \frac{1}{C}\int i_1(t) - i_2(t)\, dt$$

Aplicando LaPlace:

$$V_i(s) = LS\, I_1(s) + \frac{1}{CS}\left(I_1(s) - I_2(s)\right)$$

Malla 2:

$$\emptyset = \frac{1}{C}\int i_2(t) - i_1(t)\, dt + R\cdot i_2(t)$$

Aplicando LaPlace:

$$V_{i2}(t) = \frac{1}{CS}\left(I_2(s) - I_1(s)\right) + R\, I(s)$$

Despejamos $I_1(t)$:

$I_1(s) = (1 + RSC)\, I_2(s)$

Reemplazamos en la entrada:

$$V_i(s) = Ls([1+RSC]\, I_2(s)) + \frac{1}{Cs}\left((1+CRS)\, I_2(s) - I_2(s)\right)$$

$$V_i(s) = LS\, I_2(s) + CLS^2 R\, I_2(s) + \cancel{\frac{I_2(s)}{Cs}} + R\, I_2(s) - \cancel{\frac{I_2(s)}{Cs}}$$

$$V_i(s) = LS\, I_2(s) + CLS^2 R\, I_2(s) + R\, I_2(s)$$

$$V_i(s) = I_2(s)\,(LS + CLRS^2 + R)$$

Sabemos que: $V_o(s) = R \cdot I_2(s)$

Entonces:

$$V_i(s) = \frac{V_o(s)}{R}\,(CLRS^2 + LS + R)$$

Lo llevamos a la forma: $H(s) = \frac{\text{salida}}{\text{entrada}}$

$$\frac{V_o(s)}{V_i(s)} = \frac{R}{CLRS^2 + LS + R}$$

Quitamos la resistencia del numerador:

$$\frac{V_o(s)}{V_i(s)} = \frac{(R)\cdot\frac{1}{R}}{(CLRs^2 + Ls + R)\cdot\frac{1}{R}}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{1}{CLs^2 + \frac{Ls}{R} + 1}$$

Así, nuestra función de transferencia queda:

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{1}{CLs^2 + \frac{Ls}{R} + 1}$$